

Data Analysis: Web Scripting with Python

Web scripting helps to retrieve and analyse data from a website efficiently. In this comprehensive guide to web scraping with Python web scripting, you will learn about how to use the *requests* and *BeautifulSoup* libraries to do so.



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With the rise of web-based services, it has become mandatory for us to orient ourselves to as many of them as possible — from social networking sites to daily utilities and financial services. As these web services have become rich and reliable sources of information, businesses are using them for their analytical exercises. Web scripting is playing a critical role in web scraping. This is an essential programming technology to analyse a web page and is used to obtain information from the websites. Retrieving data from a reliable website is a common exercise for many establishments. Copying, pasting, or downloading are not practical solutions for such a job. Moreover, sometimes it becomes mandatory to process that information at regular intervals for data analysis. Web scripting is required in these circumstances, to process and analyse that data efficiently.

One industry that extensively uses scrapped data is social media marketing. Companies in this space scrape and

extract data from numerous websites to understand user sentiments, current trends/affairs, news, and feedback. This information helps to create effective marketing campaigns and business strategies.

In this article, we will explore how to perform web scraping using the *requests* library and the *BeautifulSoup* library of Python.

Python as a web scraper

A Python web scraper is a software or script to download the contents of multiple web pages and extract data from them. The functioning of a web scraper is shown in Figure 2.

Step 1: Downloading contents from web pages

In this step, a web scraper reads the requested page from a URL.

Step 2: Extracting data

Then the web scraper parses and extracts structured data from the requested contents.